

CheerUp - An Application to Boost Mood and Improve Mental Health

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Quadrant 1: Engineering Problem & Project Objectives:

Research Question:

How can I create an application that can detect the users; mood polarity and produce an applicable response containing mood-boosting advice?

Goals:

- Create a system that can help people focus on their internal state by processing their thoughts and feelings.
- Make a self-help tool to improve the users' mental health by identifying the polarity of their emotions and providing advice.
- Make a system that increases access to fast and effective mental health support and guidance which is applicable to the users' emotional needs.

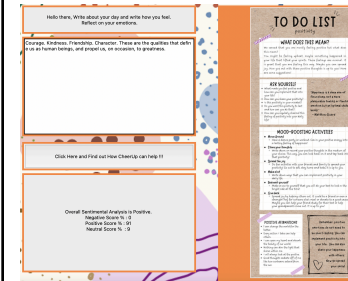
Thematic Areas:

This project is a combination of Technology and Biological Science B. More specifically, the combination of Systems Software and Behavioral and Social Sciences.

Quadrant 3: Data Analysis & Results:

Example:

I tested CheerUp with excerpts from literature. Quote from *Wonder*: "Courage. Kindness. Character. These are the qualities that define us as human beings, and propel us, on occasion, to greatness." Expected polarity: POSITIVE CheerUp result: POSITIVE



Results:

I tested 9 different excerpts and for each, I had compared the expected polarity with the Overall Sentimental Analysis Polarity that CheerUp provided. There were three excerpts from each main polarity (negative, positive, and neutral). The literature include the books *The Lion, The Witch, And The Wardrobe, The Great Gatsby, Wonder, and Make Your Bed*. CheerUp was accurate 100% of the time. This value may start to slightly vary as the number of trials increases but this proves that CheerUp is very accurate.

Quadrant 2: Project Design:

Input: textual

This input is typed by the user when they answer the prompt, "Write about your day and how you feel. Reflect on your emotions."

Push CheerUp button

The CheerUp button is what sends the input into the roBERTa system. Eventually, it will also get the data from roBERTa and send it to the Label Box.



Label Box: scores

The Label Box shows the scores that the button received back from the roBERTa model. The data includes Negative %, Positive %, Neutral %, and Overall Sentimental Analysis Polarity.

roBERTa system analysis

roBERTa is a complex transformer system that can calculate the polarity of a text. roBERTa can take into account tone, sarcasm, and context. The roBERTa system runs the text and provides the data needed.



Shows to-do list based on the main polarity

The to-do list is based on the overall sentiment analysis. The advice takes into account the varying intensity of emotion. It includes the meaning, questions to consider, a quote, mood-boosting activities, affirmations, and a content summary.

User uses the information to boost mood

Quadrant 4: Interpretation & Conclusions:

CheerUp is able to determine the polarity of the users' mood and could produce an applicable response that provided mood-boosting advice. CheerUp was proven to be extremely effective and can provide easy access to accurate emotional guidance to the user.

The model that CheerUp uses - the roBERTa model, is more accurate than some models that are used elsewhere. For example, the VADER is used on other sites and VADER scores each word (excluding stop words) and calculates the overall score. roBERTa on the other hand is a more advanced transformer model. roBERTa has the ability to take sarcasm, tone, and context into account.

Here is a comparison of CheerUp and some other similar products:

Category	CheerUp	Daylio	MoodKit
Process	- Writing a diary - Percentages of different polarities and the main polarity is detected - Advice and information is provided based on main polarity	- Choosing icons that describe mood - Choosing from activities - Given based on selection of icons	- Records daily activities - Writing a diary based on a template. - Rating of mood
Outcome	- What that result means - Questions for the user to consider - A related and motivating quote - Mood-boosting activities that are personalized based on the polarity - Affirmations that are customized based on polarity.	- Weekly reports - Reminders	- Suggestions to adopt habits - Lessons on changing thoughts - Suggestions for self-improvement